

Day 10: Normal Distribution

The Backbone of Statistics

100 Days of Data Science

Why Normal Distribution is Important

Many real-world phenomena follow a natural pattern called the normal distribution.

It appears in:

- Heights of people
- Exam scores
- Measurement errors
- Financial data

Most statistical methods assume data is approximately normal.

What is Normal Distribution?

Normal distribution is a bell-shaped curve where:

- Most values lie near the center
- Few values lie at extremes
- Data is symmetric

In a perfect normal distribution:

$$\text{Mean} = \text{Median} = \text{Mode}$$

Real-Life Example

Consider students' exam scores.

Most students score around average. Few students score very high or very low.

When plotted, scores form a bell-shaped curve.

Properties of Normal Distribution

- Symmetrical around the mean
- Mean divides curve into equal halves
- Tails extend infinitely but never touch axis

The 68-95-99.7 Rule (Empirical Rule)

In a normal distribution:

- 68% data lies within 1 standard deviation
- 95% lies within 2 standard deviations
- 99.7% lies within 3 standard deviations

Example Interpretation

If average height is 170 cm and standard deviation is 10 cm:

- Most people are between 160–180 cm
- Almost everyone is between 140–200 cm

Why It Matters in Data Science

- Many ML algorithms assume normality
- Helps in anomaly detection
- Used in hypothesis testing
- Important for confidence intervals

Normal vs Skewed Data

- Normal $\rightarrow$ balanced data
- Skewed $\rightarrow$ real-world biased data

Understanding both helps correct modeling decisions.

Common Mistakes

- Assuming all data is normal
- Ignoring skewness
- Not visualizing distribution first

Final Takeaway

Normal distribution describes how many real-world values behave.

Understanding it is essential for:

- Statistics
- Machine Learning
- Data Analysis

Most advanced techniques are built on this foundation.